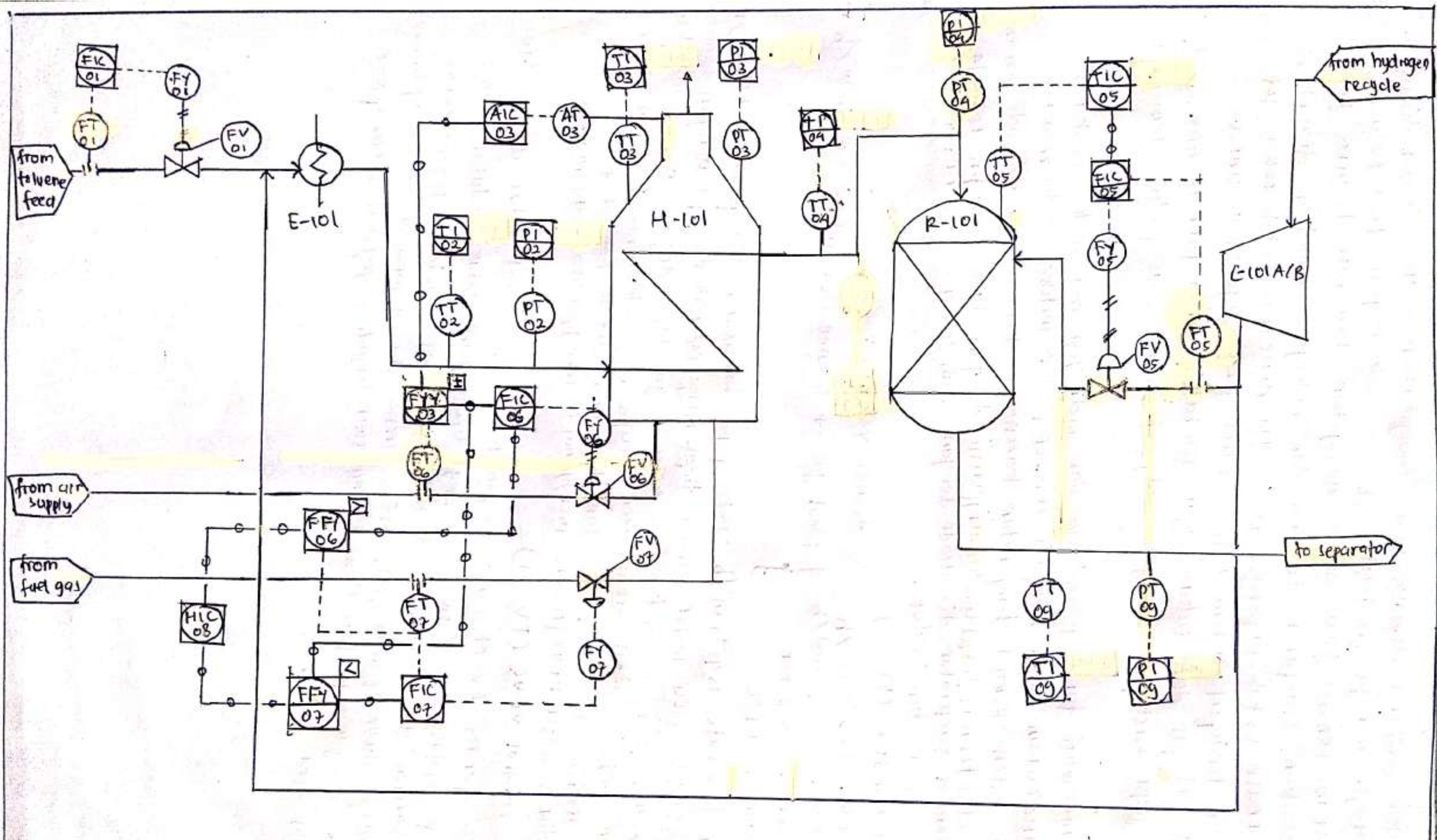


NAMA : MAULANA ISKAK
NIM : 17/413560/TK/46000

PIPING AND INSTRUMENTATION DIAGRAM. BENZENE PLANT

Proses produksi Benzena dimulai dari mengalirkan Toluena yang dicampur dengan hidrogen menuju pre-heater (E-101). Pada bagian feed toluena terdapat sistem kontrol flow untuk mengatur flow rate Toluena yang akan direaksikan. Campuran toluena dan hidrogen kemudian dialirkan menuju furnace untuk dipanaskan hingga mencapai suhu 600 °C. Pada furnace terdapat sistem ratio control yang digunakan untuk mengontrol fuel gas dan udara agar pembakaran di furnace maksimal. Selain itu juga terdapat oksigen analyzer untuk bias pada sistem rasio kontrol.

Setelah melewati furnace, toluena dan hidrogen masuk ke reaktor untuk direaksikan agar menjadi benzena dan metana. Pada reaktor terdapat sistem kontrol temperatur karena reaksi toluena menjadi benzena adalah eksotermis. Temperatur dimanipulasi dengan flow hidrogen recycle. Sistem kontrol temperatur di-cascade dengan sistem kontrol flow.

* Pemilihan sensor dan valve

a. Flow transmitter (FT-01)

mole flow : 144,2 kmol/h
mass flow : 13,3 tonne/h
Pressure : 25,8 bar
Temperature : 59 °C
flow rate : 15,3 m^3/h → dengan $\rho = 870\ kg/m^3$
: 0,15 ft^3/s
asumsi - pressure in : 27 bar
- max flow : 20 m^3/h; min flow : 10 m^3/h → CV = 196

aliran ② merupakan keluaran pompa sehingga velocity → $u = (5 + D/3)$ ft/s
didapatkan velocity, u = 5,06 ft/s
dengan diameter = 2,3 inch
≈ 3 inch

- Digunakan orifice flow meter sebagai sensor karena tekanan pipa tinggi
- Fluida mudah terbakar sehingga perlu explosion proof
- Differential pressure diukur dengan diafragma dan diubah menjadi sinyal listrik 4-20 mA untuk input kontroller (PI controller)

b. Control valve (FV-01)

- Kondisi operasi sama dengan FT-01
- Control valve dibuka dan ditutup berdasarkan flow rate yang dibutuhkan untuk produksi Benzena sehingga dibutuhkan valve yang membuka secara linier terhadap aliran fluida feed. Oleh karena itu digunakan two way linear opening valve dengan tipe fail close sehingga ketika terjadi kegagalan supply Toluene berhenti dan produksi benzena terhenti
- Control valve dikontrol dengan sinyal pneumatic
- Dengan mempertimbangkan karakteristik fluida, dipilih type body globe valve
- Fluida mudah terbakar sehingga perlu explosion proof

c. Temperature transmitter (TT-02)

Temperature : 225 °C
- Dengan temperatur 225 °C dapat digunakan RTD sebagai sensor
- Fluida mudah terbakar sehingga perlu explosion proof
- Keluaran berupa sinyal elektrik 4-20 mA untuk kemudian dikirim ke control room

d. Pressure transmitter (PT-02)

Pressure : 25,2 bar
- Digunakan gauge pressure sensor karena hanya ingin mengetahui tekanan di dalam pipa tanpa ditambah tekanan atmosfer. Selain itu gauge pressure sensor cocok digunakan untuk pipa. Gauge pressure menggunakan diafragma
- Fluida mudah terbakar sehingga perlu explosion proof
- Keluaran berupa sinyal elektrik 4-20 mA untuk kemudian dikirim ke control room

e. Temperature transmitter (TT-03)

Temperatur : 600 °C
- Dengan temperatur 600 °C maka digunakan termokopel sebagai sensor
- Furnace merupakan tempat pembakaran sehingga perlu explosion proof
- Keluaran berupa sinyal elektrik 4-20 mA untuk kemudian dikirim ke control room

f) Pressure Transmitter (PT-03)

- Pressure di furnace dapat diukur menggunakan differential pressure sensor karena volumenya yang besar. Perbedaan tekanan di bagian atas dan bawah yang terukur merupakan tekanan pada furnace.
- Keluaran berupa sinyal elektrik 4-20 mA untuk kemudian dikirim ke control room
- Furnace merupakan tempat pembakaran sehingga perlu explosion proof
- Differential pressure diukur dengan diafragma.

g) Analyzer (AT-03)

- Analyzer digunakan untuk menganalisis kandungan oksigen pada gas buang sehingga dapat diketahui apakah pembakaran terjadi secara maksimal
- Analyzer tidak ada di ISA-S20 sehingga tidak dibuat spesifikasinya
- Keluaran 4-20 mA untuk bias air flow controller dan membutuhkan explosion proof karena berada di furnace

h) Temperature Transmitter (TT-04)

- Temperature : 600 °C
- Dengan temperatur 600°C maka termokopel digunakan sebagai sensor
- Fluida mudah terbakar sehingga perlu explosion proof
- Keluaran berupa sinyal elektrik 4-20 mA dikirim ke control room

j) Pressure Transmitter (PT-04)

Pressure : 25 bar

- Digunakan gauge pressure dengan diafragma sebagai sensor karena hanya ingin mengukur tekanan dalam pipa tanpa ditambah tekanan atmosfer
- Fluida mudah terbakar sehingga perlu explosion proof
- Keluaran berupa sinyal elektrik 4-20 mA dikirim ke control room

k) Temperature Transmitter (TT-05)

Temperature : 660 °C

- Dengan temperatur 660°C maka termokopel digunakan sebagai sensor
- Terjadi reaksi eksotermis pada reaktor sehingga perlu explosion proof
- Keluaran berupa sinyal elektrik 4-20 mA dikirim ke control room, sebagai input Temperature controller

l) Flow transmitter (FT-05)

- Temperature : 41 °C
- Pressure : 25,5 bar
- mass flow : 0,36 tonne/h
- mole flow : 42,6 kmol/h
- asumsi pressure input : 26 bar
- flow rate = 43,61 m^3/h = 0,43 ft^3/s
- untuk gas flow keluaran compressor u = 20 D ft/s, didapatkan u = 5,99 ft/s dengan diameter pipa = 3,6 inch ≈ 4 inch
- asumsi flow max = 50 m^3/h } CV = 215,6 ≈ 216
- Pipa bertekanan tinggi 25,5 bar sehingga digunakan orifice flow meter
- Differential pressure diukur dengan diafragma dan dikonversi menjadi flow
- Keluaran sinyal elektrik 4-20 mA dikirim ke control room sebagai input flow controller
- Fluida pada pipa mudah terbakar sehingga perlu explosion proof

m) Flow transmitter (FT-06)
- Kondisi fluida dalam pipa tidak diketahui sehingga spesifikasi tidak dibuat
- Dapat digunakan orifice flow meter sensor untuk mengukur flow rate udara yang masuk ke furnace karena mempunyai tekanan dan flow rate tinggi.
- Keluaran berupa sinyal elektronik 4-20 mA untuk input air flow controller

n) Flow control valve (FV-06)
- Kondisi fluida dalam pipa tidak diketahui sehingga spesifikasi tidak dibuat
- Control valve dibuka dan ditutup berdasarkan air flow rate yang dibutuhkan untuk proses pembakaran sehingga dibutuhkan valve dengan bukaan linier terhadap air flow rate. Oleh karena itu digunakan two way linear opening valve dengan tipe fail close sehingga ketika terjadi kegagalan suppy udara ke furnace berhenti dan mencegah ledakan.
- Digunakan type body globe valve

o) Flow transmitter (FT-07)
- Kondisi fluida dalam pipa tidak diketahui sehingga spesifikasi tidak dibuat
- Dapat digunakan orifice flow meter sensor untuk mengukur fuel flow rate yang masuk ke furnace karena mempunyai tekanan dan flow rate yang tinggi. Dibutuhkan explosion proof karena fuel mudah terbakar
- Keluaran berupa sinyal elektronik 4-20 mA untuk input fuel flow rate

p) Flow control valve (FV-07)
- Kondisi fluida dalam pipa tidak diketahui sehingga spesifikasi tidak dibuat
- Control valve dibuka dan ditutup berdasarkan fuel flow rate yang dibutuhkan untuk proses pembakaran sehingga dibutuhkan valve dengan bukaan linier. terhadap fuel flow rate. Oleh karena itu digunakan two way linear opening valve dengan tipe fail close sehingga ketika terjadi kegagalan supply fuel ke furnace berhenti dan mencegah ledakan

q) Flow control valve (FV-05)
- Kondisi operasi sama dengan FT-05
- Control valve dibuka dan ditutup berdasarkan Hydrogen flow rate yang dibutuhkan untuk mengontrol temperatur reactor sehingga dibutuhkan valve dengan bukaan linier terhadap hydrogen flow rate. Oleh karena itu digunakan two way linear opening valve dengan tipe fail open agar reactor tidak overheat saat terjadi kegagalan

r) Temperature transmitter (TT-09)
- Temperatur : 654°C
- Dengan temperatur 659°C maka digunakan termo kopel sebagai sensor
- Fluida mudah terbakar sehingga dibutuhkan explosion proof
- Keluaran berupa sinyal elektrik 4-20 mA dikirim ke control room

s) Pressure Transmitter (PT-09)
- Tekanan : 24 bar
- Digunakan gauge pressure sensor dengan diafragma karena hanya ingin mengukur tekanan di dalam pipa tanpa ditambah tekanan atmosfer.
- Fluida mudah terbakar sehingga perlu explosion proof
- Keluaran berupa sinyal elektrik 4-20 mA dikirim ke control room

	DIFFERENTIAL PRESSURE INSTRUMENTS	SHEET 1 OF 11
		SPEC. NO. 01 / REV.
	NO / BY / DATE / REVISION	CONTRACT / DATE
		REQ. / P.O.
		BY / CHK'D / APPR.

	No.	Item	Specification
	1	Tag No.	2-FT-01 Service E101 Toluene feed
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Integ ☐ Other
	3	Case	MFR STD ☑ Nom Size ____ Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☑ Explosion proof ☐ Class
			For use in Intrinsically Safe System ☐ Other
	6	Power Supply	117V 60 Hz ☐ Other ac ____ dc ☑ 24 Volts
	7	Chart	12 in. Circ. ☐ Other ____ Range ____ No.
	8	Chart Drive	24 hr Other ____ Elec. ☐ Spring ☐ Other
	9	Scale	Type ____ Range: 1 ____ 2 ____ 3 ____
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
			For Receiver, See Spec Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate)
			Sub: s=Slow, f=Fast If☐ Df ☐ P ☐ PI ☑ PD ☐ PID ☐ I_s ☐ D_s ☐
			Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☑ Other
	15	Manual Reg.	None ☐ MFR STD ☑ Other
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
UNIT	17	Service	Flow ☑ Level ☐ Diff. Pressure ☐ Other
	18	Element Type	Diaphragm ☑ Bellows ☐ Mercury ☐ Other
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange ____ Body Rating ____ psig
	21	Diff. Range	Fixed ☑ Adj. Range ____ Set At
	22		Elevation ____ Suppression
	23	Process Data	Fluid Toluene Max Temp. ____ Max. Press.
	24	Process Conn.	½ in. NPT ☐ Other
	25	Alarm Switches	Quantity ____ Form ____ Rating
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To ____ on Inc. Meas.
	27	Options	Pressure Element ☐ Range ____ Material
			Temp. Element ☐ Range ____ Type
			Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ ____ Charts
			Valve Manifold
			Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐
			Integrator
			Other
	28	MFR & Model No.	Rosemount 3051SFC / Emerson

Notes:

ISA Form S20.20a

CONTROL VALVE DATA SHEET

ISA	PROJECT Benzene production	DATA SHEET 2 of 11
	UNIT	SPEC 02
	P.O.	TAG 2-FV-01
	ITEM	DWG
	CONTRACT	SERVICE E101 Toluene feed
	*MFR. SERIAL	

			Units	Max Flow	Norm Flow	Min Flow	Shut-Off	
1		Fluid Aromatic hydrocarbon (Toluene)				Crit Press PC		
2	SERVICE CONDITIONS	Flow Rate	m³/h	20	15,3	10	–	
3		Inlet Pressure	bar		27			
4		Outlet Pressure	bar		25,8			
5		Inlet Temperature	°C		59			
6		Spec Wt/Spec Grav/Mol Wt			0,865		–	
7		Viscosity/Spec Heats Ratio	cp		0,389		–	
8		Vapor Pressure P_V					–	
9		*Required C_V		196	196	196	–	
10		*Travel	%	100	77	100	0	
11		Allowable/*Predicted SPL	dBA	/	/	/	–	
12								

13	LINE	Pipe Line Size In 3"	53	ACTUATOR	*Type Diaphragm
14		& Schedule Out 3"	54		*Mfr & Model Emerson / Fisher 657
15		Pipe Line Insulation	55		*Size 30 Eff Area 297 cm²
16	VALVE BODY/BONNET	*Type Globe	56		On/Off Modulating
17		*Size NPS 3 ANSI Class	57		Spring Action Open/Close open
18		Max Press/Temp	58		*Max Allowable Pressure 15 psig
19		*Mfr & Model Emerson / easy e family EZ	59		*Min Required Pressure 0 psig
20		*Body/Bonnet Matl WCC steel	60		Available Air Supply Pressure:
21		*Liner Material/ID 316 L	61		Max 20 psig Min 0 psig
22		End In raised-face flanged	62		*Bench Range 3-15 psig/
23		Connection Out raised-face flanged	63		Actuator Orientation vertical
24		Flg Face Finish	64		Handwheel Type
25		End Ext/Matl	65		Air Failure Valve YES Set at 20 psig
26		*Flow Direction Flow to open	66		
27		*Type of Bonnet	67	POSITIONER	Input Signal 4-20 mA
28		Lub & iso Valve NO Lube	68		*Type single acting
29		*Packing Material PTFE	69		*Mfr & Model Emerson / Fisher 3582
30		*Packing Type Enviro-seal	70		*On Incr Signal Output Incr/Decr increase
31			71		Gauges yes By-pass yes
32	TRIM	*Type one-stage	72		*Cam Characteristic linear
33		*Size 0,5" Rated Travel 1,125"	73		
34		*Characteristic linear	74	SWITCHES	Type Quantity
35		*Balanced/Unbalanced balanced	75		*Mfr & Model
36		*Rated C_V 196 F_L 0,77 X_T 0,044	76		Contacts/Rating
37		*Plug/Ball/Disk Material S31600 W	77		Actuation Points
38		*Seat Material S31600 W	78		
39		*Cage/Guide Material CF 8M	79	AIR SET	*Mfr & Model
40		*Stem Material S31600 W	80		*Set Pressure
41			81		Filter Gauge
42			82		
43	SPECIALS/ACCESSORIES	NEC Class Group Div	83	TESTS	*Hydro Pressure
44			84		ANSI/FCI Leakage Class
45			85		
46			86		

Rev	Date	Revision	Orig	App

(Rows 47–52 of SPECIALS/ACCESSORIES blank.)

*Information supplied by manufacturer unless already specified

	TEMPERATURE INSTRUMENTS (FILLED SYSTE	SHEET 3 OF 11
		SPEC. NO. 03 / REV.
	NO / BY / DATE / REVISION	CONTRACT / DATE
		REQ. P.O.
		BY / CHK'D / APPR.

	No.	Item	Specification
	1	Tag No.	4-TT-02 Service H-101 Inlet
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size 90 CM Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☑ Explosion proof ☐ Class For Use in Intrinsically Safe System ☐ Other
	6	Power Supply	117 V 60Hz ☐ Other ac ___ dc ☑ 24 Volts
	7	Chart	Strip ☐ ___ Roll ☐ ___ Fold ☐ ___ Circular ___ Time Marks
	8	Chart Drive	Speed ___ Power
	9	Scales	Type Range 1 ___ 2 ___ 3 ___ 4
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other For Receiver See Spec. Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate), Sub: s = Slow f = Fast P ☐ PI ☐ PD ☐ PID ☐ If ☐ Df ☐ Is ☐ Ds ☐ Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other
	15	Manual Reg.	None ☐ MFR STD ☐ Other
	16	Output	4-20 mA ☐ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
ELEMENT	17	Fill	SAMA Class III A Compensation
	18	Process Data	Temp: Normal 225 °C Max ___ Max. Press.
	19	Range	Fixed ☑ Adj. Range ___ Set At Overrange Protection to
	20	Bulb	Type rigid Mtl. 321 SST Extension: Length ___ Type Size: Diameter 10 mm Length 50 mm Insertion 30 mm Conn: ___ Location ___ Ft. Above ☐ Below ☐ Instr.
	21	Capillary	MFR STD ☑ Length 50 mm Mtl. ___ Armor
	22	Well	Mtl. ___ Insertion ___ Lag Ext. ___ Conn. Const: Drilled ☑ Built-Up ☐ Other
	23	Alarm Switches	Quantity ___ Form ___ Rating
	24	Function	Temp ☐ Deviation ☐ Contacts To ___ On Temp. Increase
	25	Options	Filt-Reg. ☐ Sup. Gage ☐ Output Gage ☐ ___ Charts Other
	26	Mfr. & Model No.	Emerson / Rosemount 214C RTD

Notes:

ISA FORM S20.11a

	PRESSURE INSTRUMENTS				SHEET 4 OF 4
	NO	BY	DATE	REVISION	SPEC. NO. 04 / REV.
					CONTRACT / DATE
					REQ. / P.O.
					BY / CHK'D / APPR.

	1	Tag No. 4-PT-02	Service H 101 inlet
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size ____ Color: MFR STD ☐ Other ____
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other ____
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof☑ Class ____ For Use In Intrin. Safe System ☐ Other ____
	6	Power Supply	117V 60Hz ☐ Other ac ____ dc 24 Volts
	7	Chart	____ Strip ☐ ____ Roll ☐ ____ Fold ☐ Circular ____ Time Marks ____ Range ____ Number ____
	8	Chart Drive	Speed ____ Power ____
	9	Scales	Type ____ Range 1 ____ 2 ____ 3 ____ 4 ____
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____ For Receiver See Spec Sheet ____
CONTROLLER	11	Control Modes	P=Prop (Gain) I=Integral (Auto-Reset) D=Derivative (Rate) Sub: s=Slow f=Fast P ☐ PI ☐ PD ☐ PID ☐ If ☐ Df ☐ Is ☐ Ds ☐ Other ____
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ____
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other ____
	15	Manual Reg.	None ☐ MFR STD ☐ Other ____
	16	Output	4-20mA ☐ 10-50mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____
ELEMENT	17	Service	Gage Press. ☑ Vacuum ☐ Absolute ☐ Compound ☐
	18	Element Type	Diaphragm ☑ Helix ☐ Bourdon ☐ Bellows ☐ Other ____
	19	Material	316 SS ☐ Ber. Copper ☐ Other 316L SST
	20	Range	Fixed ☑ Adj. Range ____ Set at ____ Overrange protection to ____
	21	Process Data	Press: Normal 25,2 bar Max ____ Element Range ____
	22	Process Conn.	¼ in. NPT ☐ ½ in. NPT ☑ Other ____ Location: Bottom ☐ Back ☐ Other ____
	23	Alarm Switches	Quantity Form Rating
	24	Function	Press ☐ Deviation ☐ Contacts To ____ on Inc Press.
OPTIONS	25	Options	Filt-Reg. ☐ Sup Gage ☐ Output Gage ☐ ____ Charts Diaph Seal ☐ Type ____ Diaph ____ Bot Bowl ____ Conn ____ Capillary: Length ____ Mtl. ____ Other ____
	26	MFR & Model No.	Rosemount 2088 /Emerson

Notes:

ISA Form S20.40a

	TEMPERATURE INSTRUMENTS (FILLED SYSTEM)	SHEET 5 OF 11	
		SPEC. NO. 05	REV.
	NO | BY | DATE | REVISION	CONTRACT	DATE
		REQ. - P.O.	
		BY | CHK'D	APPR.

	No.	Item	Specification
	1	Tag No.	6-TT-04 Service R-101 inlet
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size 90 cm² Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class For Use in Intrinsically Safe System ☐ Other
	6	Power Supply	117 V 60Hz ☐ Other ac ___ dc ☑ 24 Volts
	7	Chart	Strip ☐ Roll ☐ Fold ☐ Circular Time Marks
	8	Chart Drive	Speed Power
	9	Scales	Type Range 1 2 3 4
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other For Receiver See Spec. Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate), Sub: s = Slow f = Fast P ☐ PI ☐ PD ☐ PID ☐ I_f ☐ D_f ☐ I_s ☐ D_s ☐ Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other
	15	Manual Reg.	None ☐ MFR STD ☐ Other
	16	Output	4-20 mA ☐ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
ELEMENT	17	Fill	SAMA Class III A Compensation
	18	Process Data	Temp: Normal 600 °C Max Max. Press.
	19	Range	Fixed ☑ Adj. Range Set At Overrange Protection to
	20	Bulb	Type rigid Mtl. 321 SST Extension: Length Type Size: Diameter 10 mm Length 50 mm Insertion 30 mm Conn: Location Ft. Above ☐ Below ☐ Instr.
	21	Capillary	MFR STD ☑ Length 50 mm Mtl. 321 SST Armor
	22	Well	Mtl. Insertion Lag Ext. Conn. Const: Drilled ☑ Built-Up ☐ Other
	23	Alarm Switches	Quantity Form Rating
	24	Function	Temp ☐ Deviation ☐ Contacts To ___ On Temp. Increase
	25	Options	Filt-Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts Other
	26	Mfr. & Model No.	Emerson / Rosemount 214C RTD

Notes:

ISA FORM S20.11a

		PRESSURE INSTRUMENTS	SHEET 6 OF 11
		NO | BY | DATE | REVISION	SPEC. NO. 06 | REV.
			CONTRACT | DATE
			REQ. P.O.
			BY | CHK'D | APPR.

	No.	Item	Specification
	1	Tag No. 6-PT-04	Service R101 inlet
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size ____ Color: MFR STD ☐ Other ____
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other ____
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class ____ For Use In Intrin. Safe System ☐ Other ____
	6	Power Supply	117V 60Hz ☐ Other ac ____ dc 24 Volts
	7	Chart	____ Strip ☐ ____ Roll ☐ ____ Fold ☐ Circular ____ Time Marks ____ Range ____ Number ____
	8	Chart Drive	Speed ____ Power ____
	9	Scales	Type ____ Range 1 ____ 2 ____ 3 ____ 4 ____
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____ For Receiver See Spec Sheet ____
CONTROLLER	11	Control Modes	P=Prop (Gain) I=Integral (Auto-Reset) D=Derivative (Rate) Sub: s=Slow f=Fast P ☐ PI ☐ PD ☐ PID ☐ If ☐ Df ☐ I_s ☐ D_s ☐ Other ____
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ____
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other ____
	15	Manual Reg.	None ☐ MFR STD ☐ Other ____
	16	Output	4-20mA ☐ 10-50mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____
ELEMENT	17	Service	Gage Press. ☑ Vacuum ☐ Absolute ☐ Compound ☐
	18	Element Type	Diaphragm ☑ Helix ☐ Bourdon ☐ Bellows ☐ Other ____
	19	Material	316 SS ☐ Ber. Copper ☐ Other 316L SST
	20	Range	Fixed ☑ Adj. Range ____ Set at ____ Overrange protection to ____
	21	Process Data	Press: Normal 25 bar Max ____ Element Range ____
	22	Process Conn.	¼ in. NPT ☐ ½ in. NPT ☑ Other ____ Location: Bottom ☐ Back ☐ Other ____
	23	Alarm Switches	Quantity ____ Form ____ Rating ____
	24	Function	Press ☐ Deviation ☐ Contacts To ____ on Inc Press.
OPTIONS	25	Options	Filt-Reg. ☐ Sup Gage ☐ Output Gage ☐ ____ Charts Diaph Seal ☐ Type ____ Diaph ____ Bot Bowl ____ Conn ____ Capillary: Length ____ Mtl. ____ Other ____
	26	MFR & Model No.	Emerson / Rosemount 2088

Notes:

ISA Form S20.40a

			TEMPERATURE INSTRUMENTS (FILLED SYSTEM) — NO / BY / DATE / REVISION — SHEET 7 OF 11 — SPEC. NO. 07 — REV. — CONTRACT — DATE — REQ. - P.O. — BY — CHK'D — APPR.
	1	Tag No.	R-TT-05 Service R101 benzene reactor temperature
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size 92 cm² Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☑ Explosion proof ☐ Class For Use in Intrinsically Safe System ☐ Other
	6	Power Supply	117 V 60Hz ☐ Other ac- dc☑ 24 Volts
	7	Chart	Strip ☐ Roll ☐ Fold ☐ Circular Time Marks
	8	Chart Drive	Speed Power
	9	Scales	Type Range 1 2 3 4
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other For Receiver See Spec. Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate), Sub: s = Slow f = Fast P ☐ PI ☑ PD ☐ PID ☐ If ☐ Df ☐ Is ☐ Ds ☐ Other
	12	Action	On Meas. Increase Output: Increases ☑ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☑ Other
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☑ Other
	15	Manual Reg.	None ☐ MFR STD ☐ Other
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
ELEMENT	17	Fill	SAMA Class III A Compensation
	18	Process Data	Temp: Normal 600 Max Max. Press.
	19	Range	Fixed ☑ Adj. Range Set At Overrange Protection to
	20	Bulb	Type rigid Mtl. 321 SST Extension: Length Type Size: Diameter 15 mm Length 50 mm Insertion 30 mm Conn: Location Ft. Above ☐ Below ☐ Instr.
	21	Capillary	MFR STD ☑ Length 50 mm Mtl. Armor stainless steel
	22	Well	Mtl. Insertion Lag Ext. Conn. Const: Drilled ☑ Built-Up ☐ Other
	23	Alarm Switches	Quantity Form Rating
	24	Function	Temp ☐ Deviation ☐ Contacts To On Temp. Increase
	25	Options	Filt-Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts Other
	26	Mfr. & Model No.	YOKOGAWA OS 06B01B01-00E THERMOCOUPLE

Notes:

DIFFERENTIAL PRESSURE INSTRUMENTS				SHEET 8 OF 11	
NO	BY	DATE	REVISION	SPEC. NO. 08	REV.
				CONTRACT	DATE
				REQ. - P.O.	
				BY / CHK'D	APPR.

	No.	Item	Specification
	1	Tag No.	7-FT-05 Service R101 Temperature control
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Integ ☐ Other
	3	Case	MFR STD ☑ Nom Size ____ Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class For use in Intrinsically Safe System ☐ Other
	6	Power Supply	117V 60 Hz ☐ Other ac ____ dc ☑ 24 Volts
	7	Chart	12 in. Circ. ☐ Other ____ Range ____ No.
	8	Chart Drive	24 hr Other ____ Elec. ☐ Spring ☐ Other
	9	Scale	Type ____ Range: 1 ____ 2 ____ 3
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other For Receiver, See Spec Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate) Sub: s=Slow, f=Fast If☐ Df ☐ P ☐ PI ☑ PD ☐ PID ☐ Is ☐ Ds ☐ Other
	12	Action	On Meas. Increase Output: Increases ☑ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☑ Other
	15	Manual Reg.	None ☐ MFR STD ☑ Other
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
UNIT	17	Service	Flow ☑ Level ☐ Diff. Pressure ☐ Other
	18	Element Type	Diaphragm ☑ Bellows ☐ Mercury ☐ Other
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange ____ Body Rating ____ psig
	21	Diff. Range	Fixed ☑ Adj. Range ____ Set At
	22		Elevation ____ Suppression
	23	Process Data	Fluid Hydrogen Max Temp. ____ Max. Press.
	24	Process Conn.	½ in. NPT ☐ Other
	25	Alarm Switches	Quantity ____ Form ____ Rating
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To ____ on Inc. Meas.
	27	Options	Pressure Element ☐ Range ____ Material Temp. Element ☐ Range ____ Type Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ ____ Charts Valve Manifold Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐ Integrator Other
	28	MFR & Model No.	Emerson / Rosemount 3051 S FC

Notes:

ISA Form S20.20a

ISA S20.50, Rev. 1 | **CONTROL VALVE DATA SHEET** | **Second Printing**

PROJECT Benzene production
UNIT
P.O.
ITEM
CONTRACT
•MFR. SERIAL

DATA SHEET 9 of 11
SPEC [illegible]
TAG 7-FV-05
DWG
SERVICE

1 Fluid Hydrogen — Crit Press PC

			Units	Max Flow	Norm Flow	Min Flow	Shut-Off	
2	SERVICE CONDITIONS	Flow Rate	m^3/h	50	43,61	30	—	
3		Inlet Pressure	bar		25,5			
4		Outlet Pressure	bar		26			
5		Inlet Temperature	°C		41			
6		Spec Wt/Spec Grav/Mol Wt		0,0696	0,0696	0,0696	—	
7		Viscosity/Spec Heats Ratio	cp				—	
8		Vapor Pressure P_V					—	
9		*Required C_V	[illegible]	216	216	216	—	
10		*Travel	%	100	87	0	0	
11		Allowable/*Predicted SPL	dBA	/	/	/	—	
12								

LINE
13 Pipe Line Size In 4"
14 & Schedule Out 4"
15 Pipe Line Insulation

VALVE BODY/BONNET
16 *Type Globe
17 *Size NPS 3 ANSI Class
18 Max Press/Temp
19 *Mfr & Model Emerson / easy e family EZ
20 *Body/Bonnet Matl WCC steel
21 *Liner Material/ID 316 L
22 End Connection In raised-face flanged
23 Out raised-face flanged
24 Flg Face Finish
25 End Ext/Matl
26 *Flow Direction Flow to open
27 *Type of Bonnet
28 Lub & Iso Valve NO Lube
29 *Packing Material PTFE
30 *Packing Type Enviro-steal
31

TRIM
32 *Type one-stage
33 *Size 0,5" Rated Travel 1,125"
34 *Characteristic linear
35 *Balanced/Unbalanced balanced
36 *Rated C_V 216 F_L 0,87 x_T 0,019
37 *Plug/Ball/Disk Material S[illegible]600 W
38 *Seat Material S[illegible]600 W
39 *Cage/Guide Material CF 8M
40 *Stem Material S31600 W
41
42

SPECIALS/ACCESSORIES
43 NEC Class Group Div
44
45
46
47
48
49
50
51
52

ACTUATOR
53 *Type Diaphragm
54 *Mfr & Model Emerson / Fisher 657
55 *Size 30 Eff Area 297 cm^2
56 On/Off Modulating
57 Spring Action Open/Close open
58 *Max Allowable Pressure 15 psig
59 *Min Required Pressure 0 psig
60 Available Air Supply Pressure:
61 Max 20 psig Min 0 psig
62 *Bench Range 3-15 psig /
63 Actuator Orientation vertical
64 Handwheel Type
65 Air Failure Valve yes Set at 20 psig
66

POSITIONER
67 Input Signal 4-20 mA
68 *Type single acting
69 *Mfr & Model Emerson / Fisher 3582
70 *On Incr Signal Output Incr/Decr increase
71 Gauges yes By-pass yes
72 *Cam Characteristic linear
73

SWITCHES
74 Type Quantity
75 *Mfr & Model
76 Contacts/Rating
77 Actuation Points
78

AIR SET
79 *Mfr & Model
80 *Set Pressure
81 Filter Gauge
82

TESTS
83 *Hydro Pressure
84 ANSI/FCI Leakage Class
85
86

Rev	Date	Revision	Orig	App

*Information supplied by manufacturer unless already specified

TEMPERATURE INSTRUMENTS (FILLED SYSTEM)				SHEET 10 OF 11	
NO	BY	DATE	REVISION	SPEC. NO. 10	REV.
				CONTRACT	DATE
				REQ. - P.O.	
				BY / CHK'D	APPR.

	No.	Item	Specification
	1	Tag No.	9-TT-09 Service R101 outlet
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size 98 cm² Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class For Use in Intrinsically Safe System ☐ Other
	6	Power Supply	117 V 60Hz ☐ Other ac ___ dc ☑ 24 Volts
	7	Chart	Strip ☐ Roll ☐ Fold ☐ Circular Time Marks
	8	Chart Drive	Speed Power
	9	Scales	Type Range 1 2 3 4
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other For Receiver See Spec. Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate), Sub: s = Slow f = Fast P ☐ PI ☐ PD ☐ PID ☐ If ☐ Df ☐ Is ☐ Ds ☐ Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other
	15	Manual Reg.	None ☐ MFR STD ☐ Other
	16	Output	4-20 mA ☐ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
ELEMENT	17	Fill	SAMA Class III A Compensation
	18	Process Data	Temp: Normal 654 Max Max. Press.
	19	Range	Fixed ☑ Adj. Range Set At Overrange Protection to
	20	Bulb	Type rigid Mtl. 321 SST Extension: Length Type Size: Diameter 15 mm Length 50 mm Insertion 30 mm Conn: Location Ft. Above ☐ Below ☐ Instr.
	21	Capillary	MFR STD ☑ Length 50 mm Mtl. Armor
	22	Well	Mtl. Insertion Lag Ext. Conn. Const: Drilled ☑ Built-Up ☐ Other
	23	Alarm Switches	Quantity Form Rating
	24	Function	Temp ☐ Deviation ☐ Contacts To ___ On Temp. Increase
	25	Options	Filt-Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts Other
	26	Mfr. & Model No.	YOKOGAWA GS 06B01B01-00E THERMOCOUPLE

Notes:

ISA FORM S20.11a

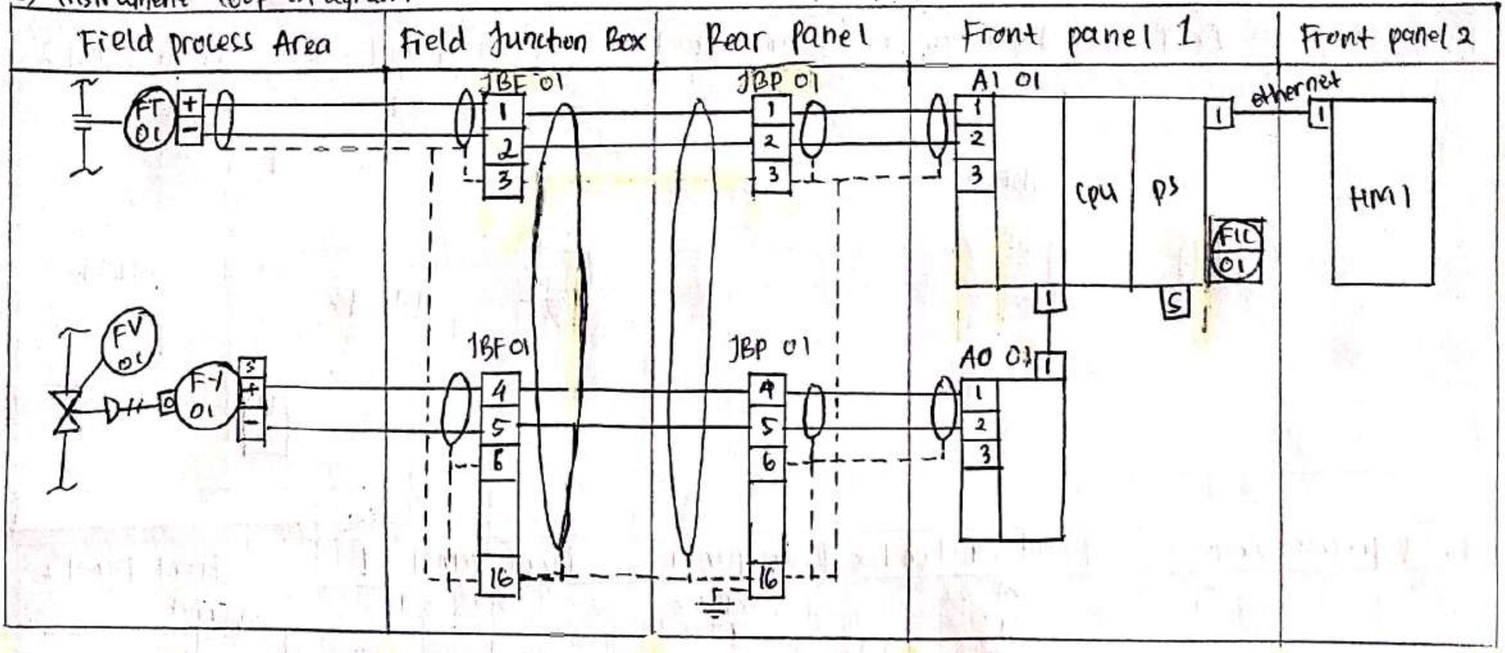
	PRESSURE INSTRUMENTS			SHEET 11 OF 11
	NO	BY	DATE	REVISION
				SPEC. NO. 11 / REV.
				CONTRACT / DATE
				REQ. / P.O.
				BY / CHK'D / APPR.

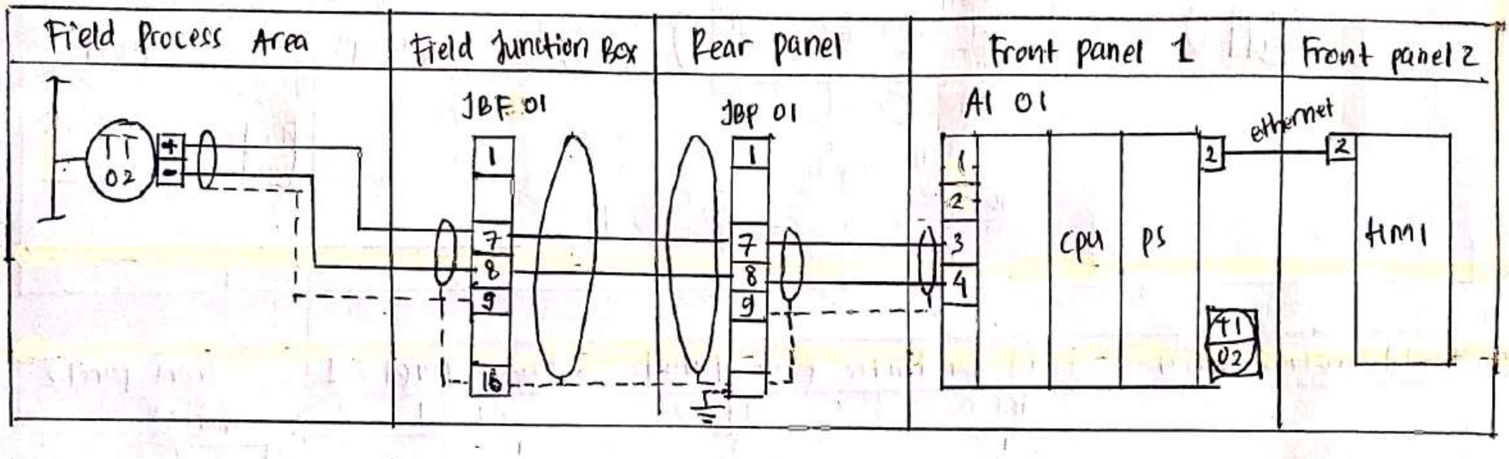
	1	Tag No.	9-PT-09 Service R101 outlet
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size ____ Color: MFR STD ☐ Other ____
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other ____
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class ____ For Use In Intrin. Safe System ☐ Other ____
	6	Power Supply	117V 60Hz ☐ Other ac ____ dc 24 Volts
	7	Chart	____ Strip ☐ ____ Roll ☐ ____ Fold ☐ Circular ____ Time Marks ____ Range ____ Number ____
	8	Chart Drive	Speed ____ Power ____
	9	Scales	Type ____ Range 1 ____ 2 ____ 3 ____ 4 ____
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____ For Receiver See Spec Sheet ____
CONTROLLER	11	Control Modes	P=Prop (Gain) I=Integral (Auto-Reset) D=Derivative (Rate) Sub: s=Slow f=Fast P ☐ PI ☐ PD ☐ PID ☐ If ☐ Df ☐ Is ☐ Ds ☐ Other ____
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ____
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other ____
	15	Manual Reg.	None ☐ MFR STD ☐ Other ____
	16	Output	4-20mA ☐ 10-50mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____
ELEMENT	17	Service	Gage Press. ☑ Vacuum ☐ Absolute ☐ Compound ☐
	18	Element Type	Diaphragm ☑ Helix ☐ Bourdon ☐ Bellows ☐ Other ____
	19	Material	316 SS ☐ Ber. Copper ☐ Other 316L SST
	20	Range	Fixed ☑ Adj. Range ____ Set at ____ Overrange protection to ____
	21	Process Data	Press: Normal 24 bar Max ____ Element Range ____
	22	Process Conn.	¼ in. NPT ☐ ½ in. NPT ☑ Other ____ Location: Bottom ☐ Back ☐ Other ____
	23	Alarm Switches	Quantity ____ Form ____ Rating ____
	24	Function	Press ☐ Deviation ☐ Contacts To ____ on Inc Press.
OPTIONS	25	Options	Filt-Reg. ☐ Sup Gage ☐ Output Gage ☐ ____ Charts Diaph Seal ☐ Type ____ Diaph ____ Bot Bowl ____ Conn ____ Capillary: Length ____ Mtl. ____ Other ____
	26	MFR & Model No.	Emerson / Rosemount 2088

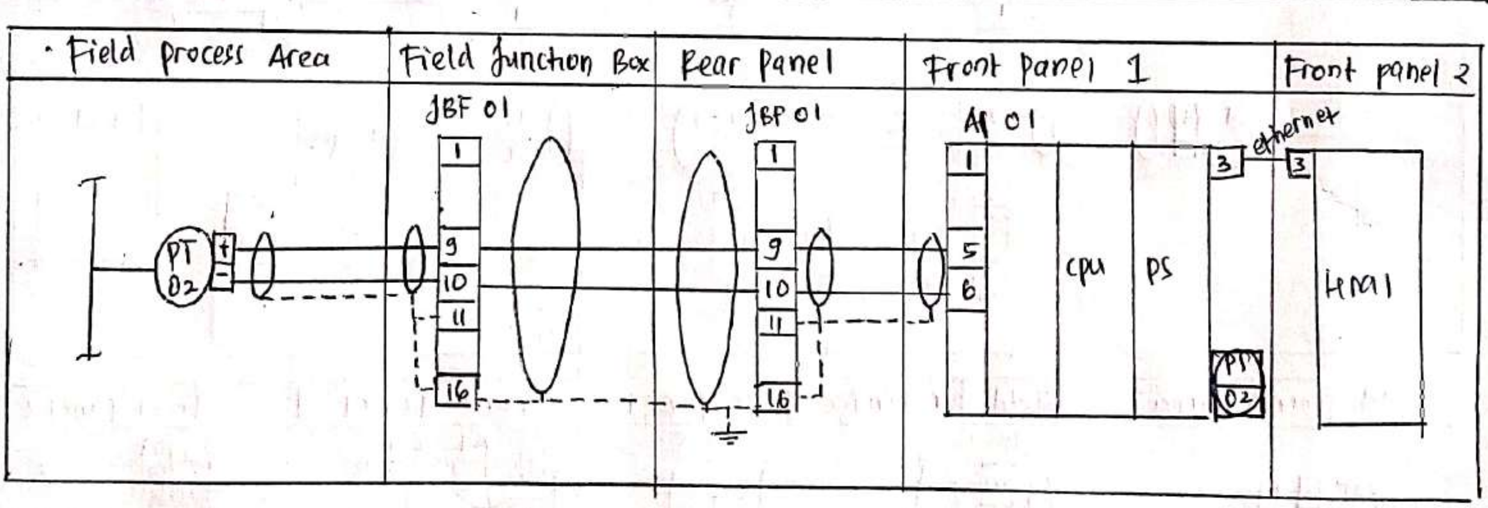
Notes:

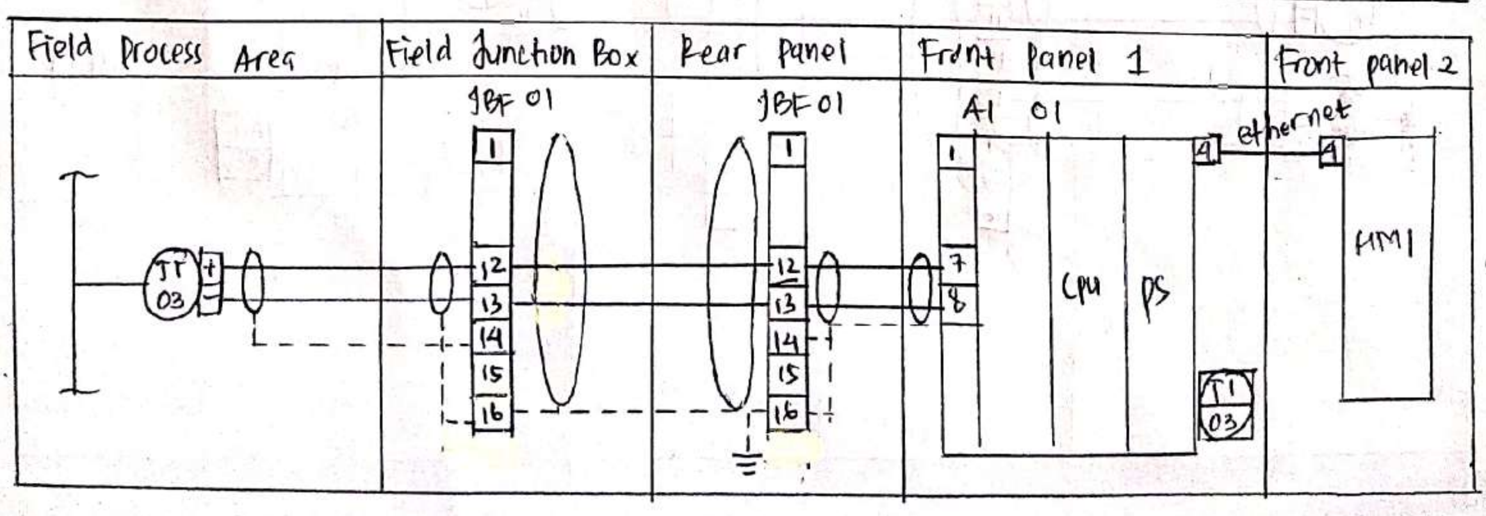
ISA Form S20.40

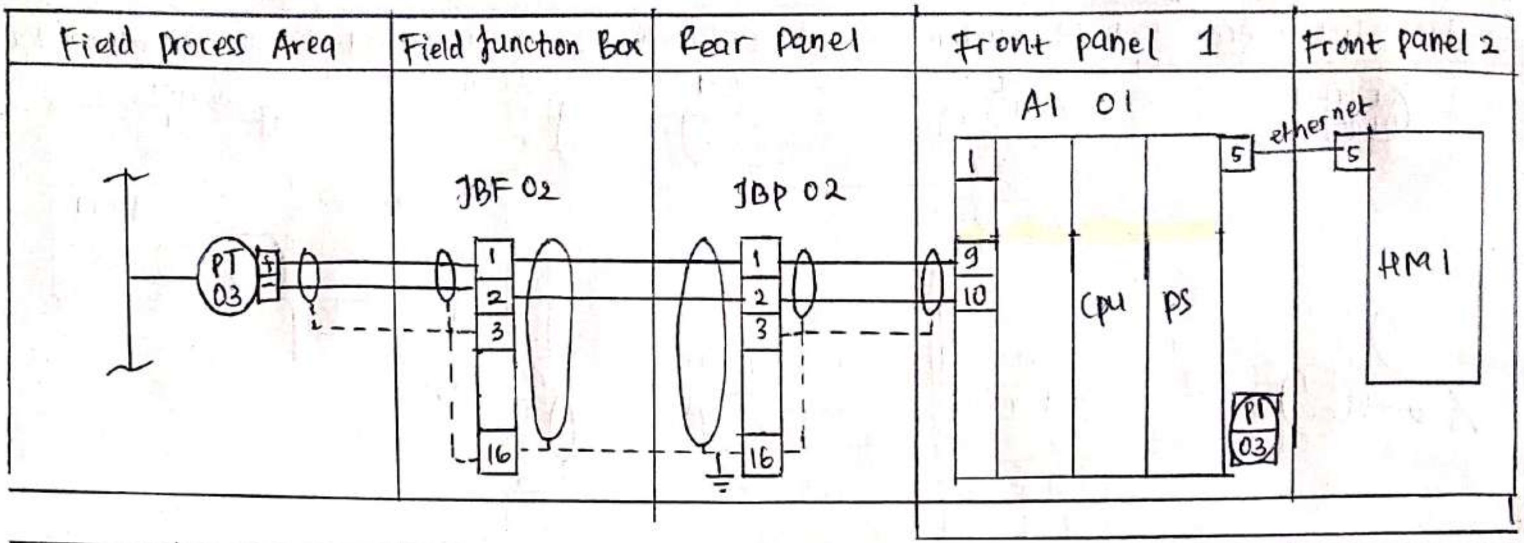
8) instrument loop diagram

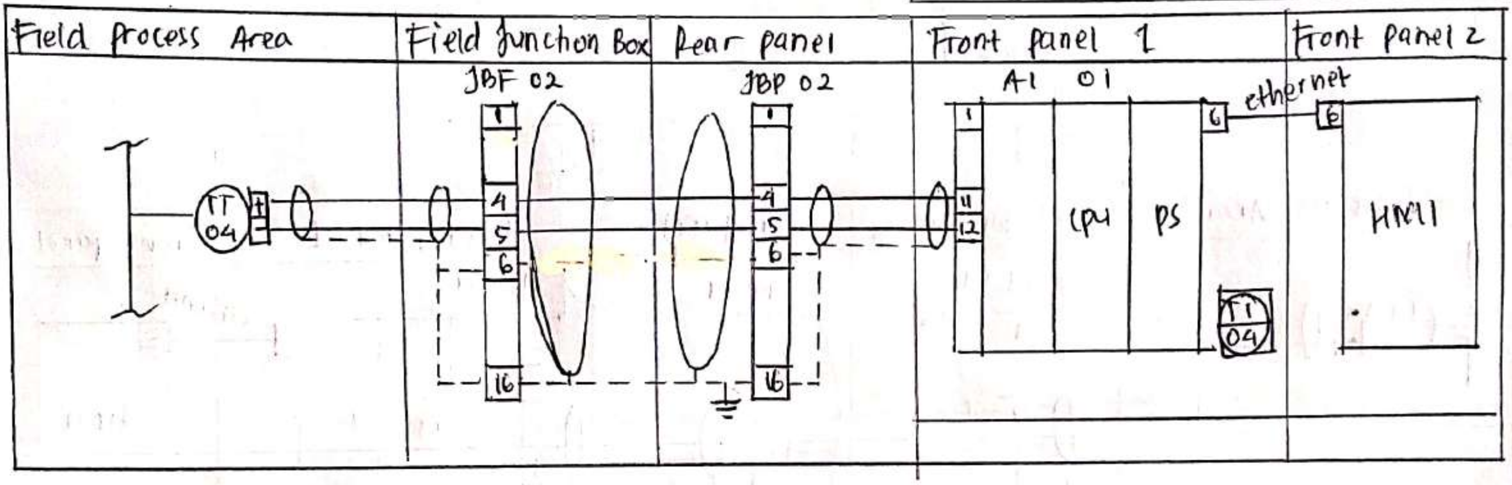


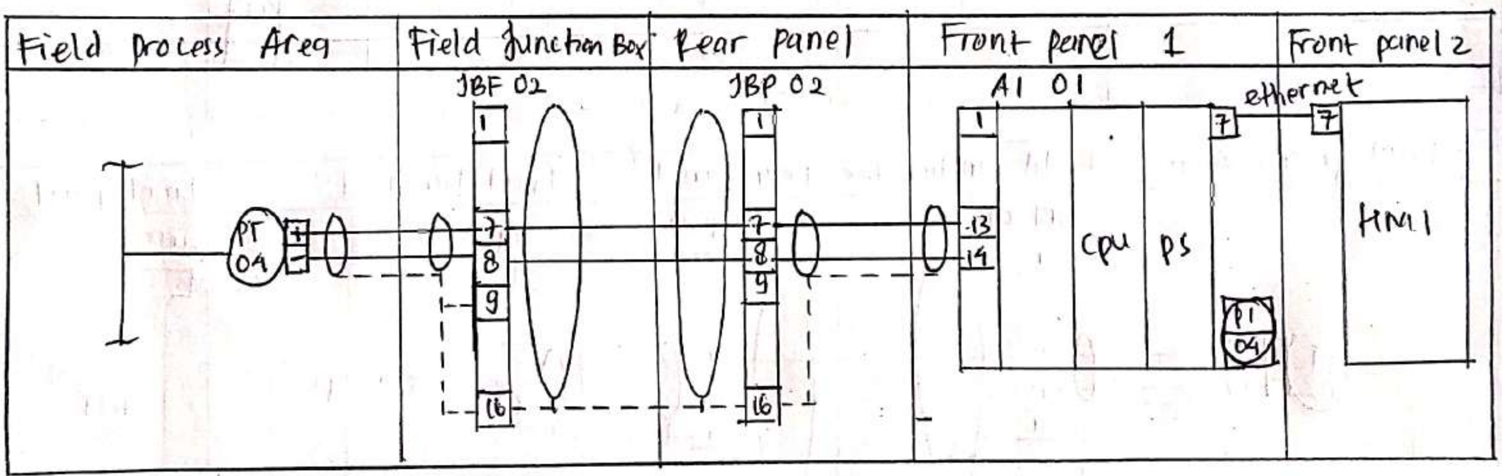


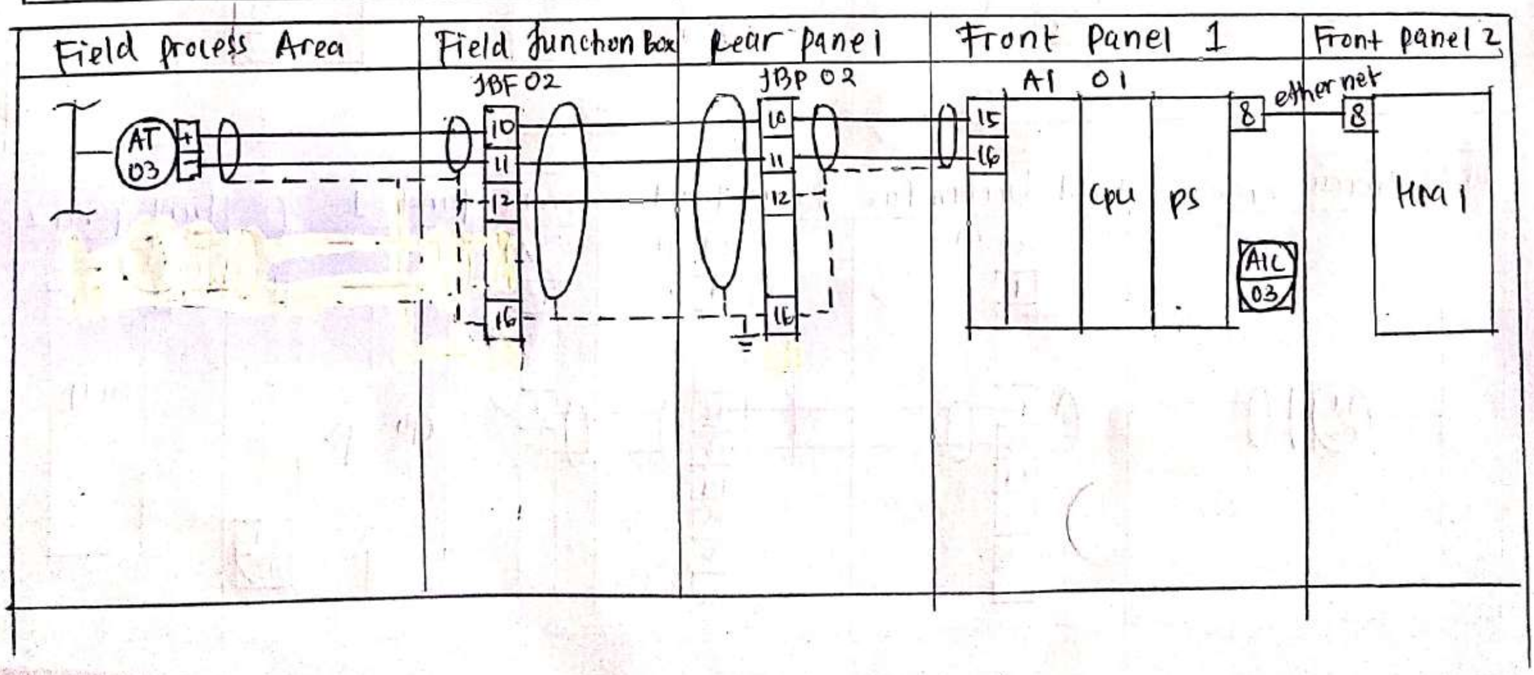





Field process Area
Field Junction Box
Rear panel
Front panel 1
Front panel 2
AI 01
ethernet
5
5
1
JBF 02
JBP 02
PT 03
+
-
1
2
3
16
1
2
3
16
9
10
cpu
ps
HMI
PT 03


Field process Area
Field Junction Box
Rear panel
Front panel 1
Front panel 2
JBF 02
JBP 02
AI 01
ethernet
6
6
1
TT 04
+
-
4
5
6
16
4
5
6
16
11
12
cpu
ps
HMI
TT 04


Field process Area
Field Junction Box
Rear panel
Front panel 1
Front panel 2
JBF 02
JBP 02
AI 01
ethernet
7
7
1
PT 04
+
-
7
8
9
16
7
8
9
16
13
14
cpu
ps
HMI
PT 04


Field process Area
Field Junction Box
Rear panel
Front Panel 1
Front panel 2
JBF 02
JBP 02
AI 01
ethernet
8
8
AT 03
+
-
10
11
12
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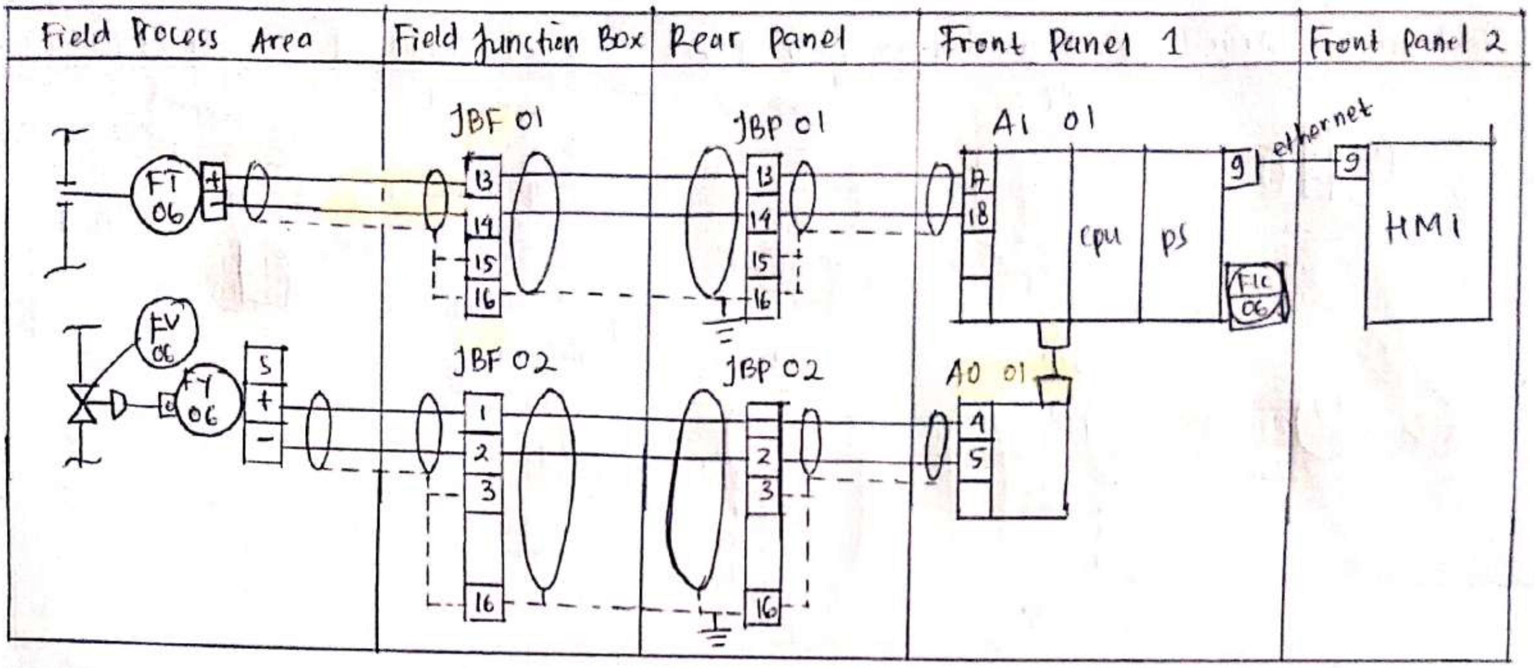
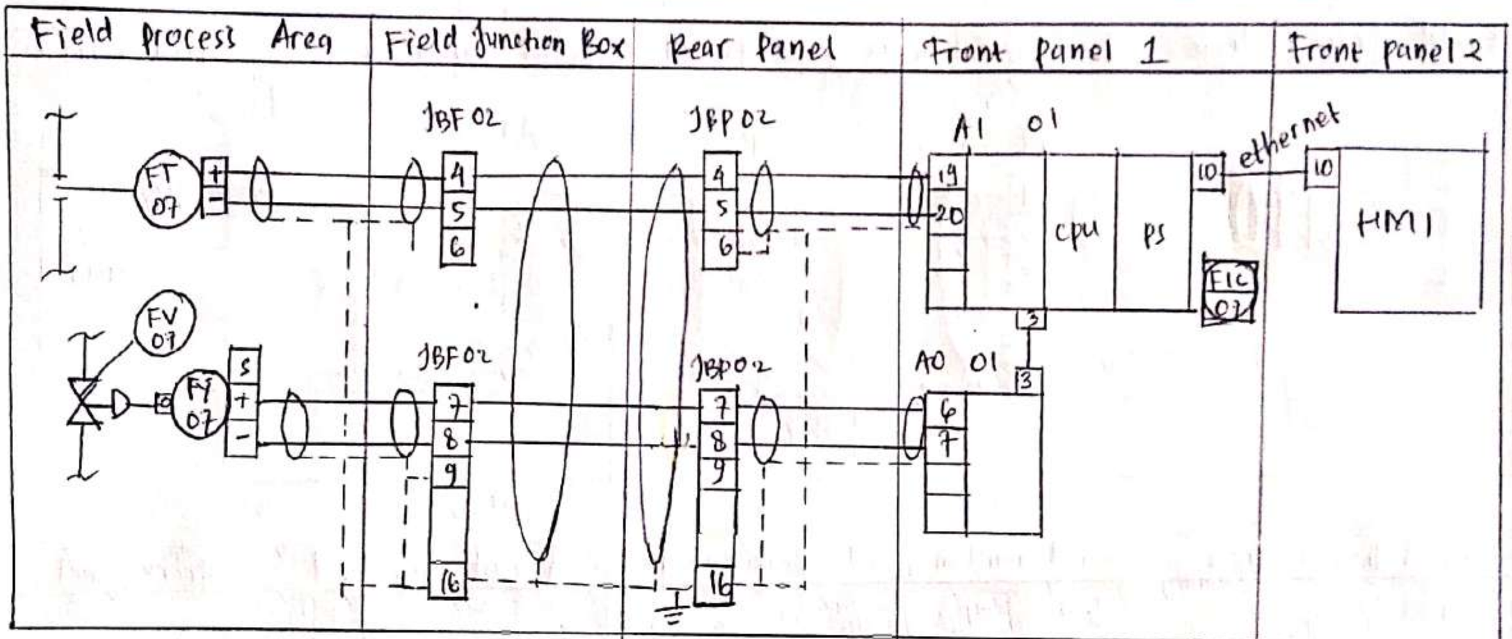
16
cpu
ps
HMI
AIC 03


Field Process Area
Field Junction Box
Rear Panel
Front Panel 1
Front panel 2
JBF 01
JBP 01
AI 01
ethernet
FT 06
13
14
15
16
17
18
cpu
ps
9
HMI
FIC 06
FV 06
FY 06
S
+
-
JBF 02
JBP 02
AO 01
1
2
3
4
5

Field Process Area
Field Junction Box
Rear Panel
Front panel 1
Front panel 2
JBF 02
JBP 02
AI 01
ethernet
FT 07
4
5
6
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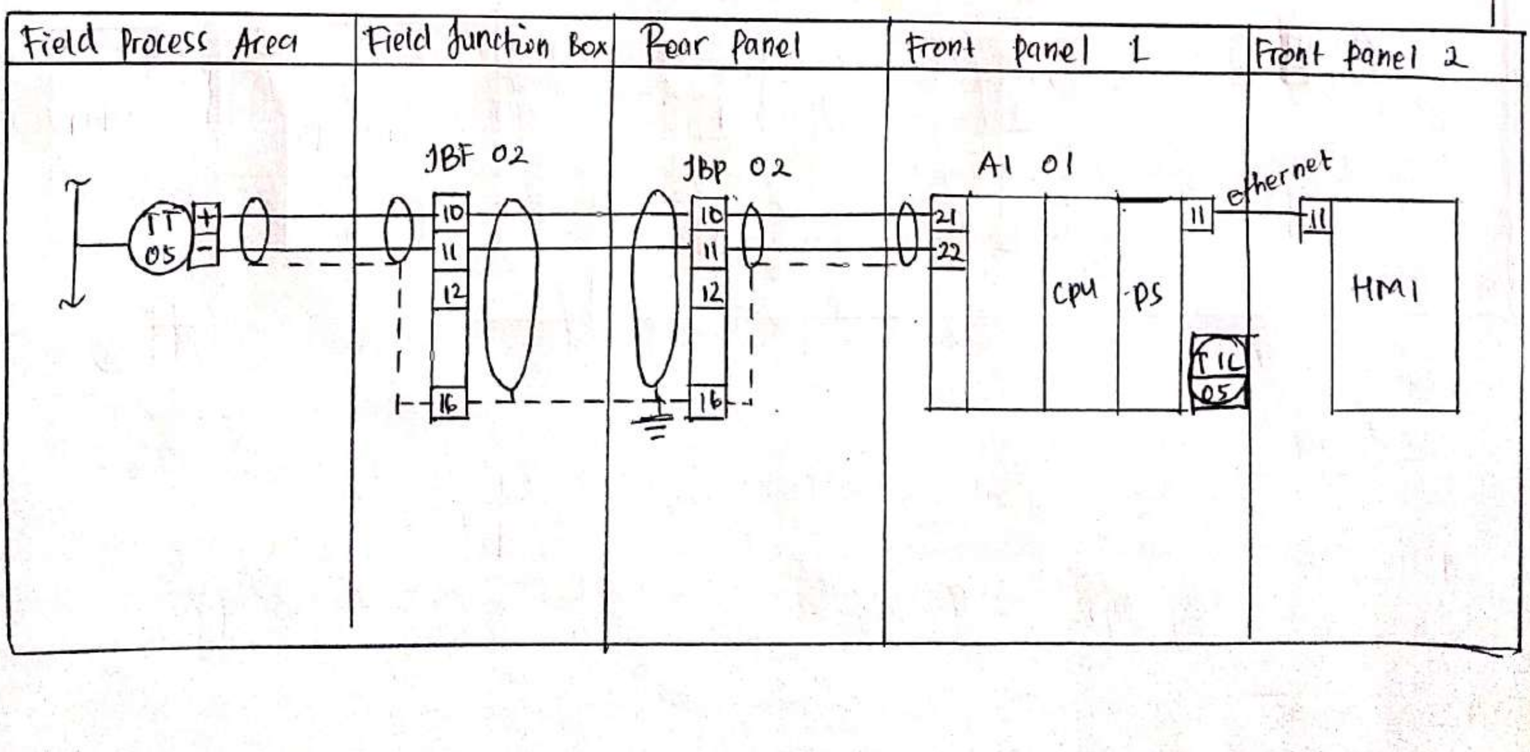
7
8
9
16
3

Field Process Area
Field Junction Box
Rear Panel
Front panel 1
Front panel 2
JBF 02
JBP 02
AI 01
ethernet
TT 05
10
11
12
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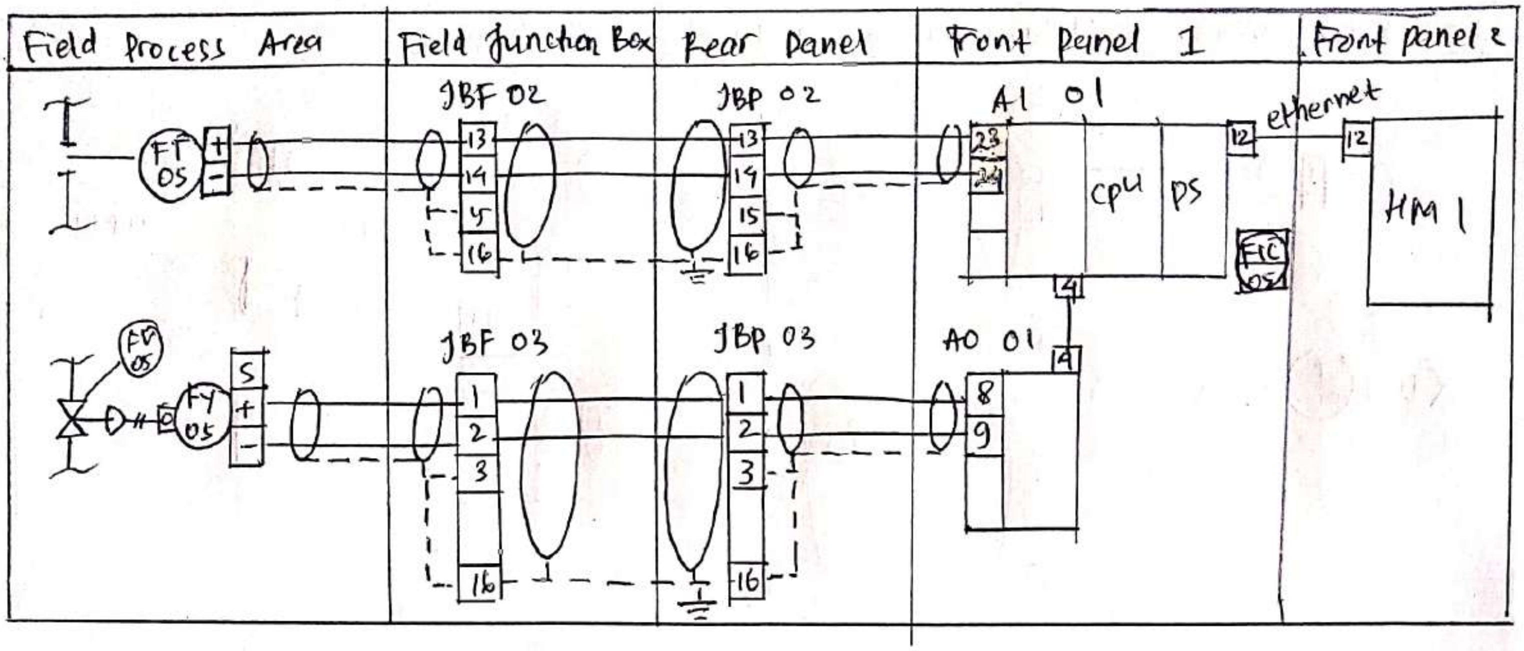
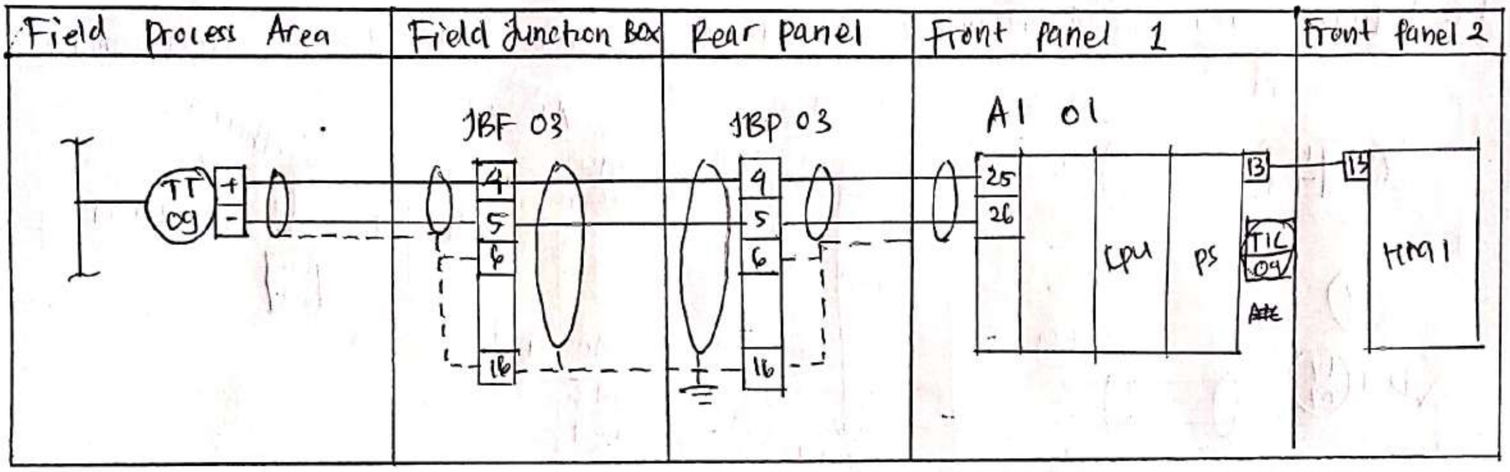
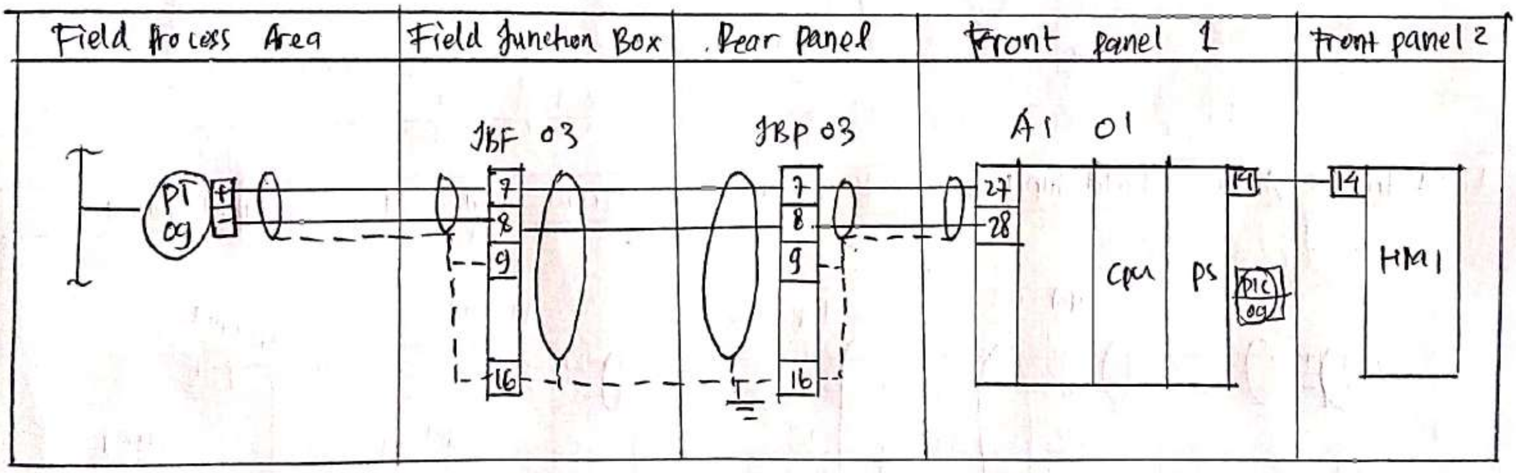
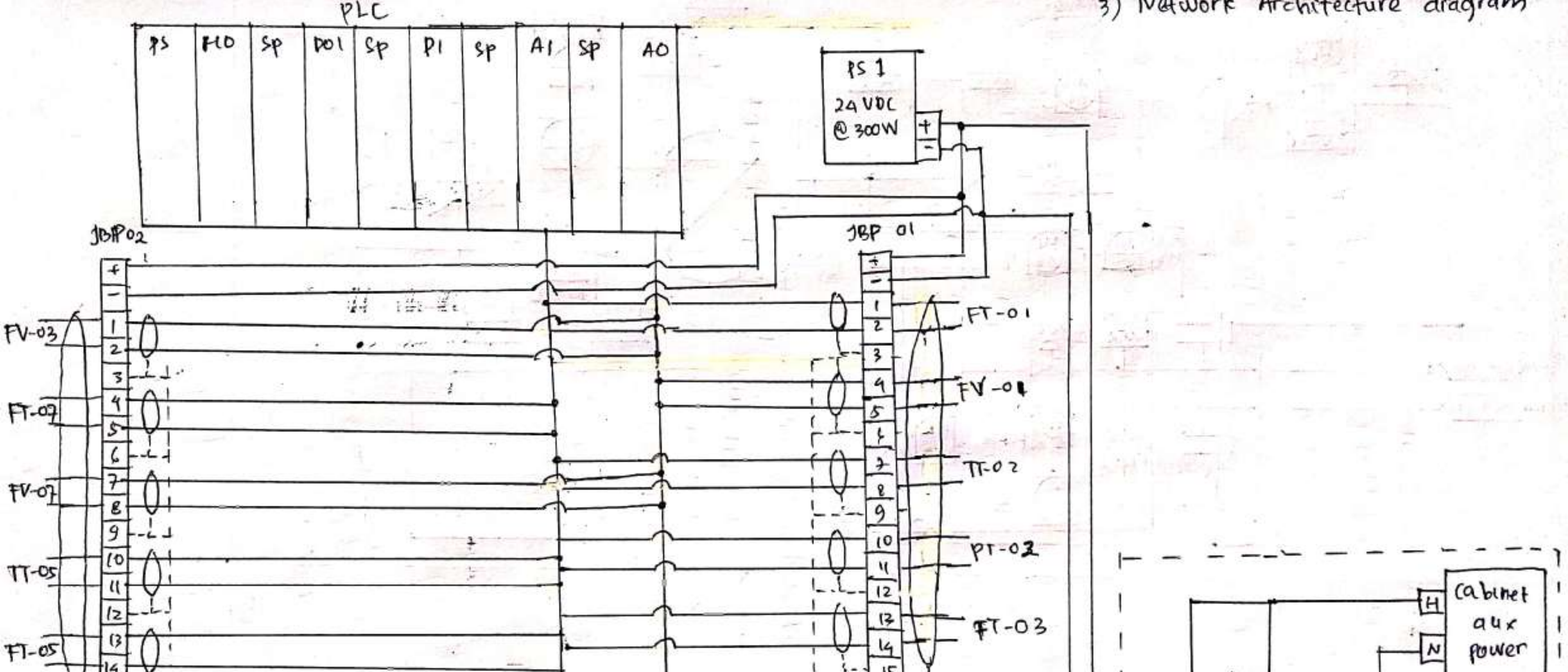
ps
11
HMI
TIC 05


Field Process Area
Field Junction Box
Rear Panel
Front Panel 1
Front panel 2
JBF 02
JBP 02
AI 01
ethernet
FT 05
13
14
15
16
23
24
CPU
PS
12
HMI 1
FIC 05
4
FV 05
FY 05
S
+
−
JBF 03
JBP 03
AO 01
1
2
3
8
9

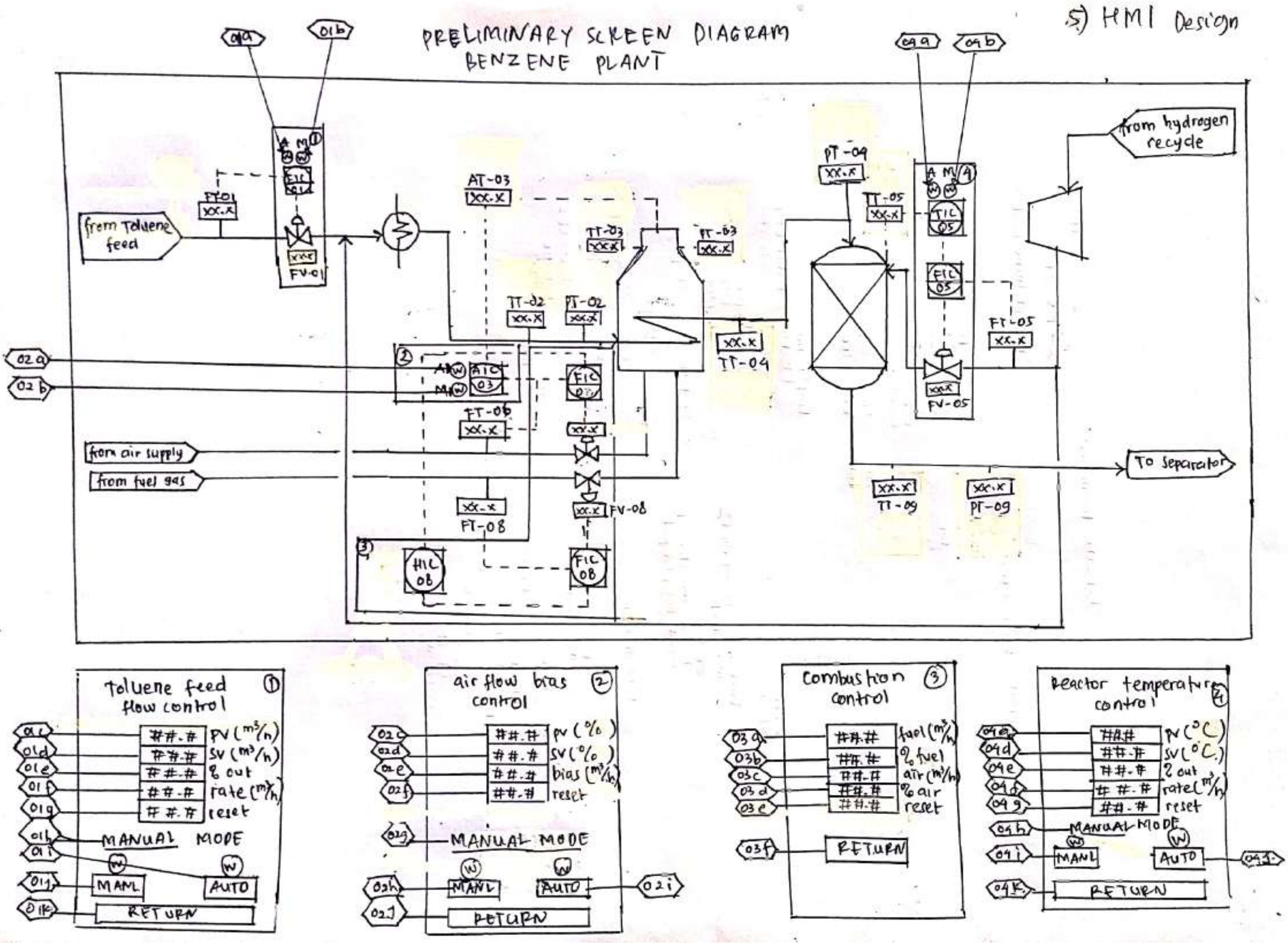

Field Process Area
Field Junction Box
Rear panel
Front panel 1
Front panel 2
JBF 03
JBP 03
AI 01
TT 09
+
−
4
5
6
16
25
26
CPU
PS
13
TIC 09
HMI


Field Process Area
Field Junction Box
Rear panel
Front panel 1
Front panel 2
JBF 03
JBP 03
AI 01
PT 09
+
−
7
8
9
16
27
28
CPU
PS
14
PIC 09
HMI

3) Network Architecture diagram



5) HMI Design


PRELIMINARY SCREEN DIAGRAM
BENZENE PLANT
from Toluene feed
FT-01
FV-01
AT-03
TT-03
PT-03
TT-02
PT-02
TT-04
PT-04
TT-05
TIC 05
FIC 05
FT-05
FV-05
from hydrogen recycle
AIC 03
FT-06
from air supply
from fuel gas
FT-08
FV-08
HIC 08
FIC 08
To Separator
TT-09
PT-09
Toluene feed flow control
PV (m³/h)
SV (m³/h)
% out
rate (m³/h)
reset
MANUAL MODE
MANL
AUTO
RETURN
air flow bias control
pv (%)
SV (%)
bias (m³/h)
reset
MANUAL MODE
MANL
AUTO
RETURN
Combustion control
fuel (m³/h)
% fuel
air (m³/h)
% air
reset
RETURN
Reactor temperature control
PV (°C)
SV (°C)
% out
rate (m³/h)
reset
MANUAL MODE
MANL
AUTO
RETURN

Node	Screen	Item	Description	Device	Action tag	Action	Animation tag	Animation
CR-1	B-OVIEW	01 a	Auto mode indicator	Lamp	na	na	T_auto	white
CR-1	B-OVIEW	01 b	manual mode indicator	Lamp	na	na	T_manl	white
CR-1	B-OVIEW	01 c	process value flow	numeric disp	na	na	T-pv	value
CR-1	B-OVIEW	01 d	set value flow	numeric disp	na	na	T_sv	value
CR-1	B-OVIEW	01 e	% flow valve	numeric disp	na	na	T_%	value
CR-1	B-OVIEW	01 f	Toluene valve flow rate	numeric disp	na	na	T_rate	value
CR-1	B-OVIEW	01 g	reset flow. value	numeric dis.	na	na	T_reset	value
CR-1	B-OVIEW	01 h	Mode message	message	na	na	T_msg	"manual.."/"auto..."
CR-1	B-OVIEW	01 i	auto mode button	mom.PB	auto_T	auto mode	na	na
CR-1	B-OVIEW	01 j	manual mode button	mom.PB	manual_T	manual mode	na	na
CR-1	B-OVIEW	01 k	return button	mom.PB	return_T	return	na	na
CR-1	B-OVIEW	02 a	auto mode indicator	lamp	na	na	O-auto	white
CR-1	B-OVIEW	02 b	manual mode indicator	lamp	na	na	O-manl	white
CR-1	B-OVIEW	02 c	process value oxygen	num.dis.	na	na	O_pv	value
CR-1	B-OVIEW	02 d	set value oxygen	num.dis.	na	na	O_SV	value
CR-1	B-OVIEW	02 e	bias air flow	num.dis.	na	na	O-bias	Value
CR-1	B-OVIEW	02 f	reset oxygen value	num.dis.	na	na	O_reset	value
CR-1	B-OVIEW	02 g	mode Message	message	na	na	O_msg	"manual.."/"auto.."
CR-1	B-OVIEW	02 h	manual mode button	mom.PB	auto-O	auto mode	na	na
CR-1	B-OVIEW	02 i	auto mode button	mom.PB	manual-O	manual mode	na	na
CR-1	B-OVIEW	02 j	return button	mom.PB	return-O	return	na	na
CR-1	B-OVIEW	03 a	fuel flow needed	num.dis.	na	na	F_flow	value
CR-1	B-OVIEW	03 b	% fuel flow valve	num.dis.	na	na	F_%	Value
CR-1	B-OVIEW	03 c	air flow rate	num.dis.	na	na	A-flow	value
CR-1	B-OVIEW	03 d	% air flow valve	num.dis	na	na	A-%	value
CR-1	B_OVIEW	03 e	fuel flow reset	num.dis	na	na	F_reset	value
CR-1	B-OVIEW	03 f	return button	mom.PB	return-F	return	na	na
CR-1	B-OVIEW	04 a	auto mode indicator	lamp	na	na	H-auto	white
CR-1	B-OVIEW	04 b	manual mode indicator	lamp	na	na	H-manl	white
CR-1	B-OVIEW	04 c	process value temperature	num.dis	na	na	H-pv	value
CR-1	B-OVIEW	04 d	set value temperature	num.dis	na	na	H-SV	value
CR-1	B-OVIEW	04 e	% flow valve	num.dis	na	na	H-%	value
CR-1	B-OVIEW	04 f	hydrogen valve flow rate	num.dis	na	na	H_rate	value
CR-1	B-OVIEW	04 g	reset flow value	num.dis	na	na	H.reset	value
CR-1	B-OVIEW	04 h	mode message	message	na	na	H-msg	"manual.."/"auto.."
CR-1	B-OVIEW	04 i	auto mode button	mom.PB	auto-H	auto mode	na	na
CR-1	B-OVIEW	04 j	manual mode button	mom.PB	manual-H	manual mode	na	na
CR-1	B-OVIEW	04 k	return button	mom.PB	return-H	return	na	na